PRE-MODELLING EVIDENCE AUDIT

# Dominance, hierarchical partitioning and group-aware feature selection

A training-only methods and results report for the frozen rare missense-SNV modelling cohort.

| | |
|---|---|
| Cohort | 39,552 retained variants from a frozen 40,000-variant source |
| Training analysis | 27,727 variants across 5,206 genes |
| Resampling | Five gene-grouped folds and 5,000 paired gene-bootstrap samples |
| Primary audit model | Balanced logistic regression with fold-local preprocessing |
| Protection | Validation and test outcomes were not used |

This document explains what was tested, why the methods were suitable, how the figures should be read and what the results do - and do not - support.

Prepared from the committed pre-modelling evidence audit outputs

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## How to read the report

The subset-performance figures compare every candidate combination inside one evidence block. The dominance figures distribute the block's total cross-validated MCC gain among its features. The group-aware figures then test whether rankings and subset choices remain stable when entire genes are held out. These views are complementary and should not be treated as competing final-model votes.

# 1. Executive summary

Two connected analyses were completed before formal modelling. Exact dominance analysis quantified how much cross-validated classification performance was associated with each candidate feature across every possible within-block subset. Group-aware feature selection tested whether those conclusions remained stable across unseen genes and whether simpler subsets remained within prespecified MCC-loss margins.

A single, fixed logistic-regression audit model was used throughout. This prevented model-family or hyperparameter changes from being confused with feature effects. The analysis was restricted to the training partition and preserved the frozen cohort and locked gene-separated split.

| Evidence block | Central training-only finding | Governance interpretation |
|---|---|---|
| Conservation | phyloP contributed most. PhastCons added supported value. The extra all-three versus phyloP + PhastCons gain was 0.00132 MCC with an interval crossing zero. | Primary biological evidence; the audit itself left the final two-versus-three decision unresolved. |
| Amino-acid properties | The block gain was small. Hydropathy ranked first, while the pair and single-feature alternatives were unstable across folds. | Biologically related measurements; removal was not finalised by the audit. |
| Gene constraint | Missense Z and LOEUF contributed similarly. The pair was selected in all five outer folds, and removing either produced a supported loss. | Retain as one grouped gene-constraint representation in the provisional audit interpretation. |
| Population frequency | Maximum-population AF outperformed global AF. Adding global AF after maximum-population AF supplied no stable gain. | Sensitivity-only because coverage was 59.75% and missingness was class-associated. |
| Computational predictors | REVEL and AlphaMissense contributed most; CADD added supported value. SIFT and PolyPhen-2 added little beyond the stronger trio. | Comparator or sensitivity evidence only; performance cannot resolve provenance and circularity concerns. |

No validation-confirmation command was executed, the test partition remained locked, and no final thesis model was trained during these analyses.

# 2. Data boundary and leakage safeguards

The analysis began from the already frozen and audited modelling assets. No variant was regenerated, removed, rebalanced or reassigned. The 40,000-row source reconciled to 39,552 retained variants and 448 preserved gene-mapping exclusions. Existing train, validation and test assignments contained 27,727, 5,790 and 6,035 variants respectively, with zero gene overlap and zero variant overlap.

| Control | Implementation |
|---|---|
| Training-only access | Feature associations, subset comparisons, ranking and methodological decisions used only the 27,727-row training partition. |
| Gene grouping | All variants from one reconciled gene were assigned to the same cross-validation fold. |
| Fold-local preprocessing | Median imputation and standard scaling were fitted only on fitting-fold rows. |
| Missing evidence | Missing biological measurements were retained as missing and were never converted to zero. |
| Held-out protection | Validation outcomes were not inspected for feature selection; test outcomes were not inspected. |
| Audit reproducibility | The analysis specification, fold assignment, random seed, outputs and file hashes were preserved. |

## Why gene grouping mattered

Gene-level attributes such as Missense Z and LOEUF are shared by every variant mapped to the same gene. A random variant split could therefore expose the same gene-level information during both fitting and evaluation. Grouped folds imposed a more demanding test: each held-out prediction was produced by a model that had not been fitted on variants from that gene.

# 3. Evidence blocks and controlled ML design

| Block | Candidate features | Subsets | Fixed background |
|---|---|---|---|
| Conservation | GERP++, phyloP, PhastCons | 8 | 46 other approved primary features |
| Amino-acid properties | Hydropathy delta, polarity delta | 4 | 47 other approved primary features |
| Gene constraint | Missense Z, LOEUF | 4 | 47 other approved primary features |
| Population sensitivity | Global AF, maximum-population AF | 4 | All 49 primary biological features |
| Predictor sensitivity | SIFT, PolyPhen-2, REVEL, CADD, AlphaMissense | 32 | All 49 primary biological features |

## Fixed audit model

Each subset was evaluated with logistic regression using C=1, the liblinear solver, balanced class weights, a maximum of 2,000 iterations and seed 20261018. A probability threshold of 0.5 was used for binary classification. Pooled out-of-fold MCC was the primary value function; PR-AUC, ROC-AUC and Brier score were retained as supporting metrics.

Logistic regression was suitable as a controlled audit instrument because the outcome was binary, probability estimates were required, and the fixed linear model enabled transparent subset comparisons. It was not presented as the final thesis model and may not capture every nonlinear interaction.

| | |
|---|---|
| 1 | Choose one evidence block and hold the remaining governed features fixed. |
| 2 | Enumerate all 2^p candidate subsets. |
| 3 | Fit fold-local imputation, scaling and logistic regression in five gene-grouped folds. |
| 4 | Collect one out-of-fold probability for every training variant. |
| 5 | Calculate pooled MCC and supporting metrics for every subset. |
| 6 | Decompose MCC gain and resample genes to quantify uncertainty. |

# 4. Dominance and hierarchical partitioning

For a candidate subset S, the value v(S) was defined as the pooled out-of-fold MCC produced by the fixed background plus S. For each feature, the MCC change obtained by adding that feature was calculated in every context where it was absent. Exact Shapley weights then averaged those marginal changes across all possible feature orders.

The resulting raw contribution represented conditional predictive contribution under the fixed audit model. The implementation verified the efficiency identity: the sum of all feature contributions equalled the full-subset MCC minus the empty-subset MCC to a tolerance of 1e-10. When that total gain was positive, normalised shares were calculated and required to sum to 100%.

Normalised shares were accounting quantities, not probabilities that a feature was necessary. They did not establish biological independence or causality. A correlated feature could receive a material average dominance share while adding almost nothing after a stronger preferred subset was already present.

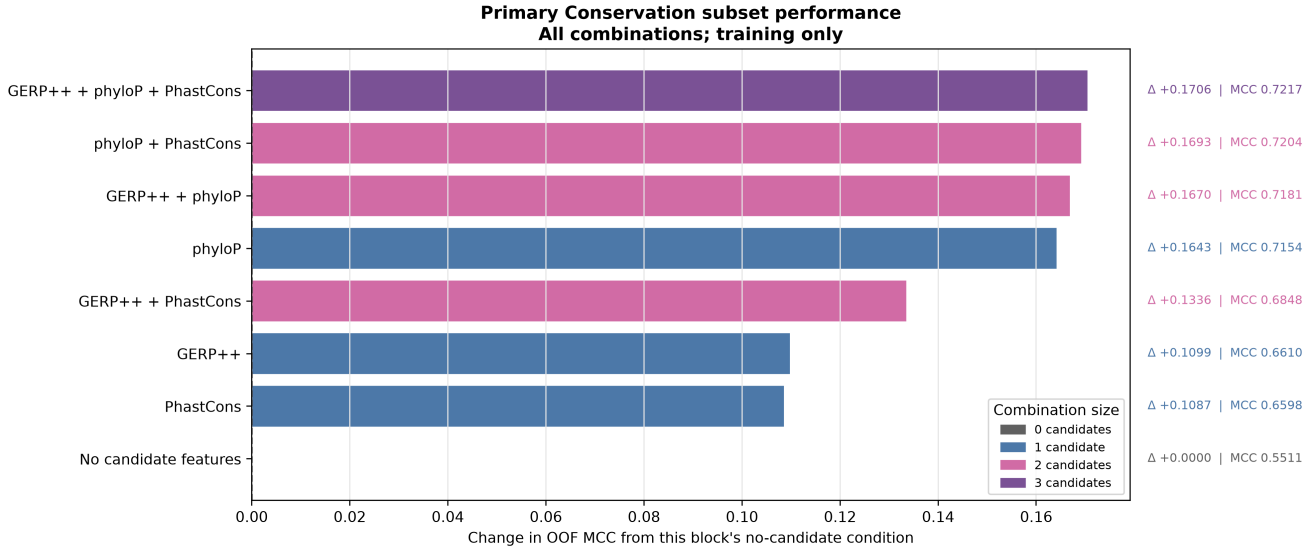
## Uncertainty estimation

Five thousand deterministic paired bootstrap samples were generated by resampling genes with replacement. All variants belonging to a sampled gene were retained with the same multiplicity, and the same resample was applied to every subset in a block. MCC values, Shapley contributions and paired full-versus-reduced differences were recalculated for every sample.

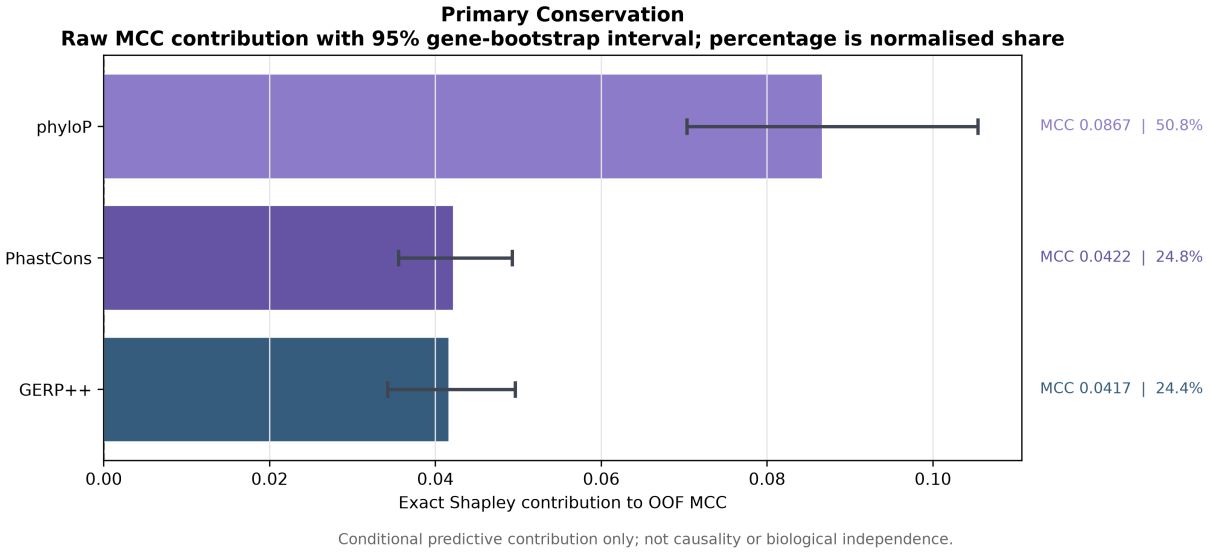
This design preserved within-gene dependence and yielded bootstrap medians, percentile 95% intervals and probabilities that predefined paired differences were greater than zero.

# 5.1 Conservation evidence

Eight conditions were tested from no conservation candidate through all three scores. The fixed background contained the remaining 46 approved primary features.




Subset-performance view for 5.1 conservation evidence. Bars show the change in pooled OOF MCC relative to the same block's no-candidate condition. Raw MCC values are printed on the figure.

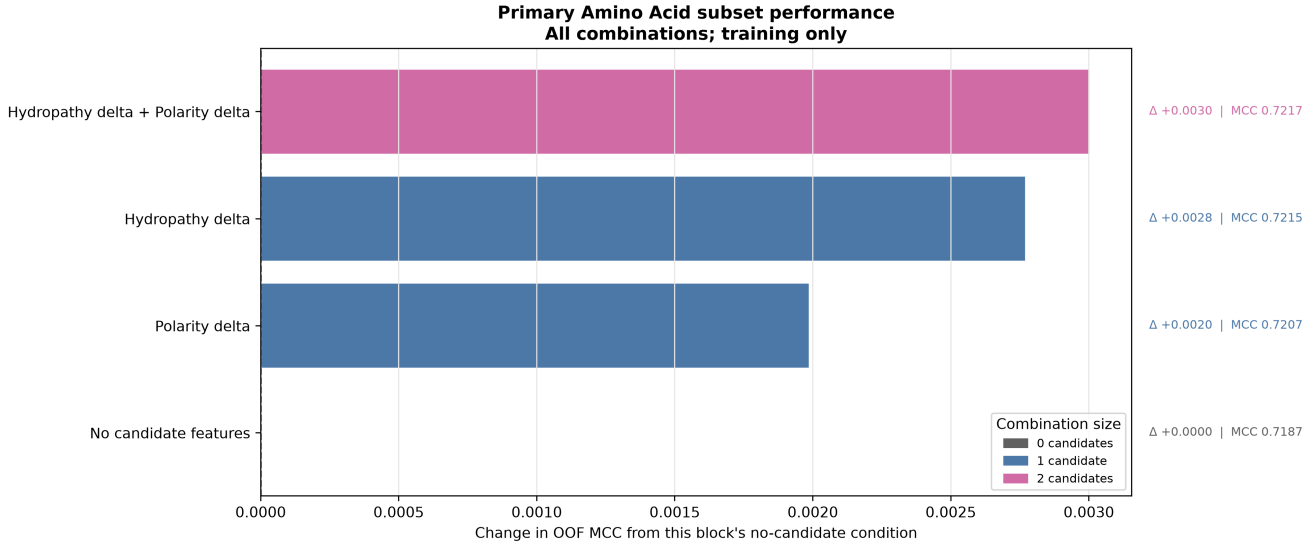



Dominance view for 5.1 conservation evidence. Bars show exact Shapley/general-dominance contributions to OOF MCC; whiskers show 95% gene-bootstrap intervals. Percentages are normalised within the block and do not imply causality.

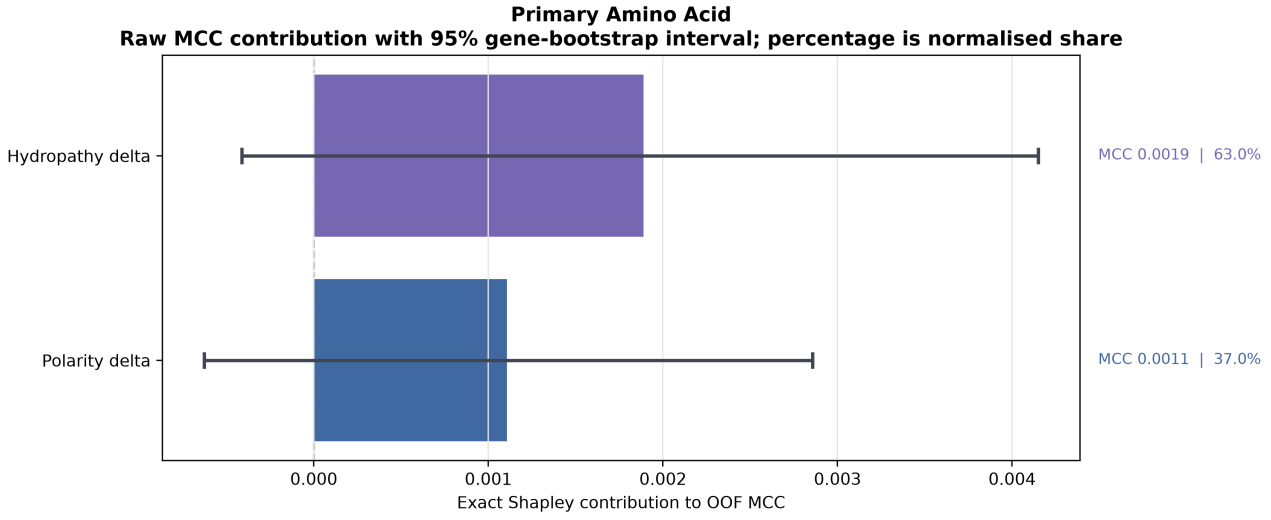
Interpretation. The empty condition produced MCC 0.55112. phyloP alone reached 0.71541; phyloP + PhastCons reached 0.72041; all three reached 0.72173. phyloP received 50.8% of the block's dominance gain. The all-three minus phyloP + PhastCons difference was 0.001315, with a 95% interval from -0.000772 to 0.003411. The incremental GERP++ gain was therefore small and uncertain in the final two-versus-three comparison.

# 5.2 Amino-acid property evidence

Four conditions were tested: neither candidate, hydropathy delta, polarity delta and both candidates. The fixed background contained the remaining 47 approved primary features.




Subset-performance view for 5.2 amino-acid property evidence. Bars show the change in pooled OOF MCC relative to the same block's no-candidate condition. Raw MCC values are printed on the figure.

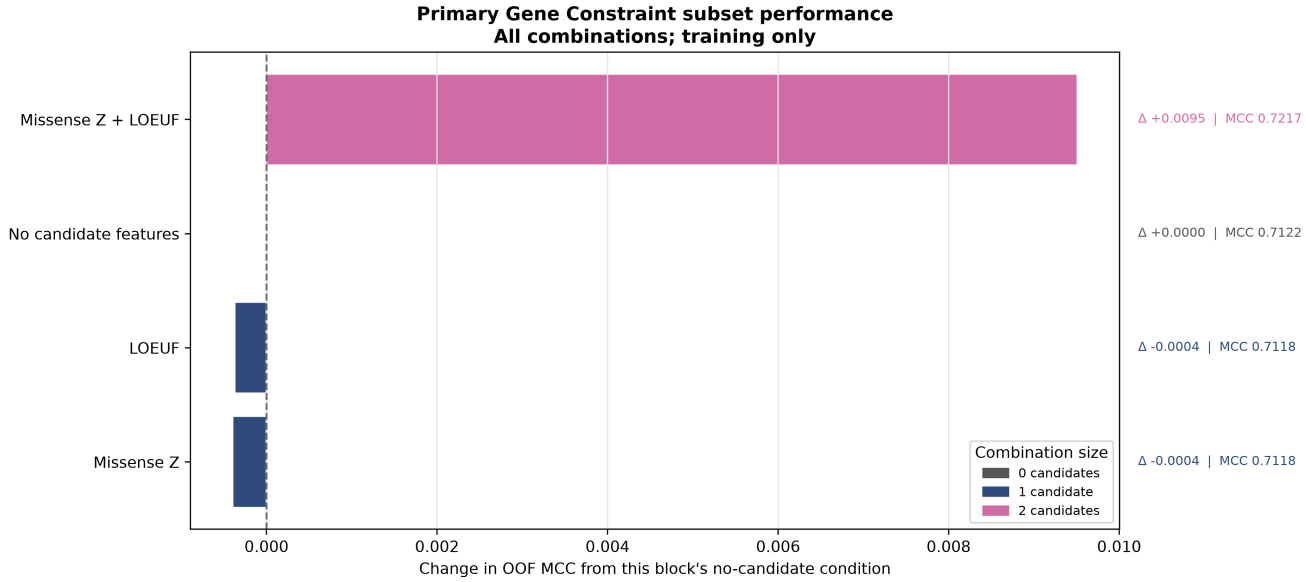



Dominance view for 5.2 amino-acid property evidence. Bars show exact Shapley/general-dominance contributions to OOF MCC; whiskers show 95% gene-bootstrap intervals. Percentages are normalised within the block and do not imply causality.

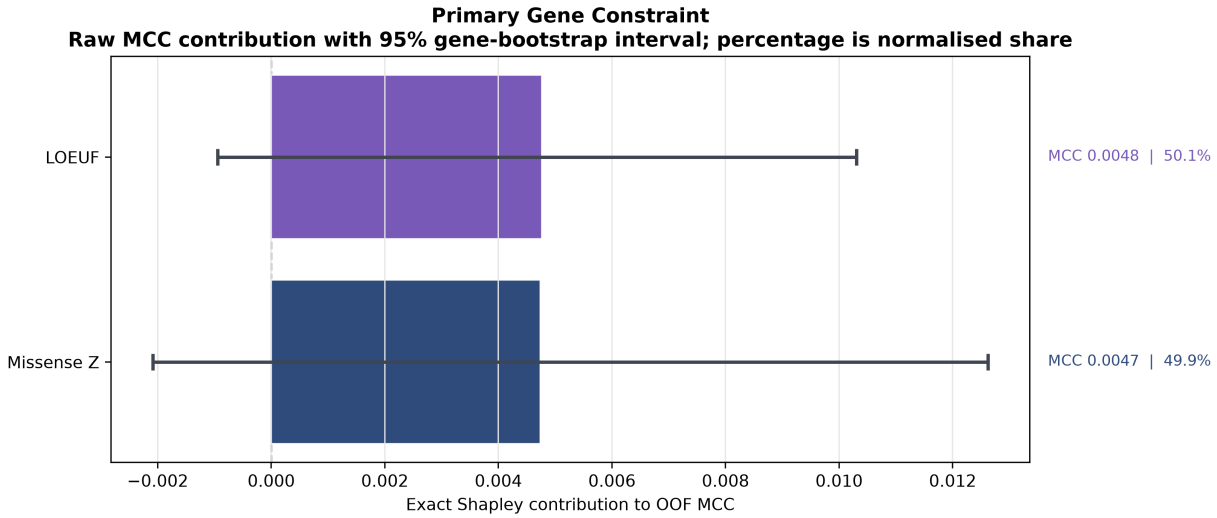
Interpretation. The full pair improved MCC by 0.003005 over the block's empty condition. Hydropathy received 63.0% of that gain and polarity received 37.0%, but both bootstrap contribution intervals included zero. Adding polarity to hydropathy changed MCC by only 0.000234, with an interval crossing zero. The audit therefore treated the reduced and full representations as unstable rather than declaring a definitive removal.

# 5.3 Gene-constraint evidence

Four conditions were tested: neither candidate, Missense Z, LOEUF and the pair. The fixed background contained the remaining 47 approved primary features.




Subset-performance view for 5.3 gene-constraint evidence. Bars show the change in pooled OOF MCC relative to the same block's no-candidate condition. Raw MCC values are printed on the figure.

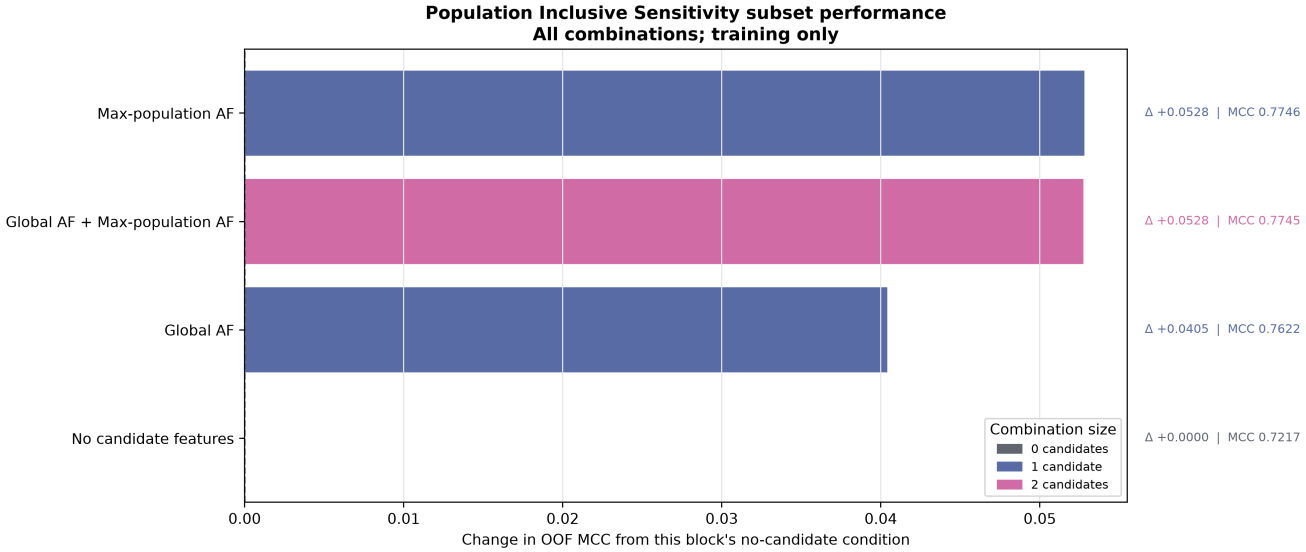



Dominance view for 5.3 gene-constraint evidence. Bars show exact Shapley/general-dominance contributions to OOF MCC; whiskers show 95% gene-bootstrap intervals. Percentages are normalised within the block and do not imply causality.

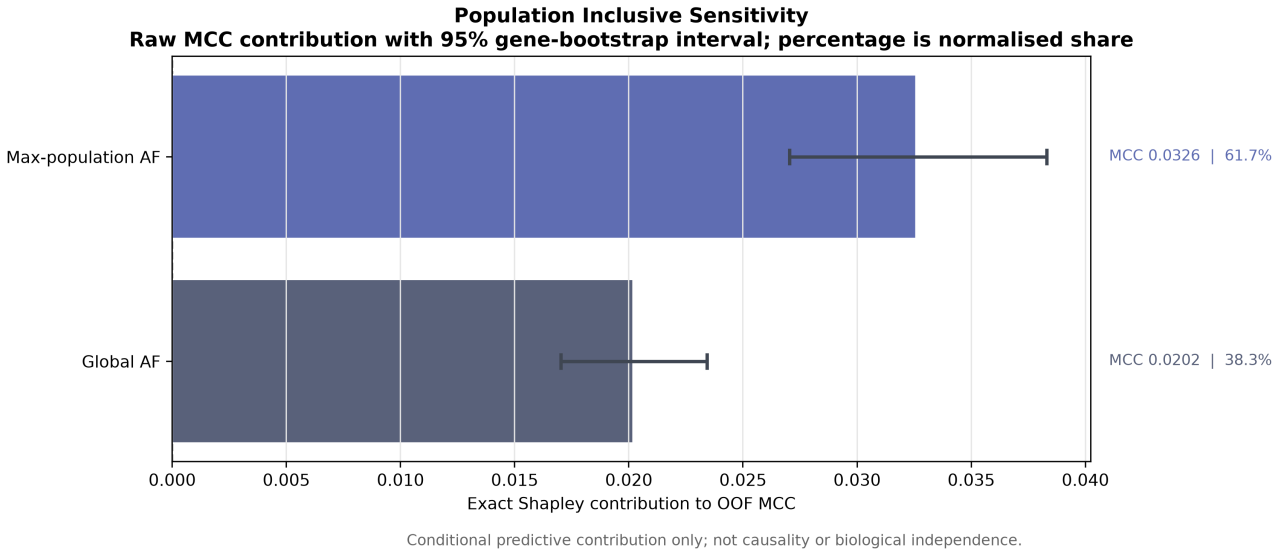
Interpretation. The pair improved MCC by 0.009509 over the empty condition. Dominance shares were nearly equal: 49.9% for Missense Z and 50.1% for LOEUF. Removing either feature from the pair produced an MCC loss near 0.0099 with a bootstrap interval above zero. The two singleton models were almost indistinguishable, but neither replaced the pair reliably.

# 5.4 Population-frequency sensitivity

Four sensitivity conditions were tested on top of all 49 primary biological features. Missing frequency was never interpreted as zero.




Subset-performance view for 5.4 population-frequency sensitivity. Bars show the change in pooled OOF MCC relative to the same block's no-candidate condition. Raw MCC values are printed on the figure.

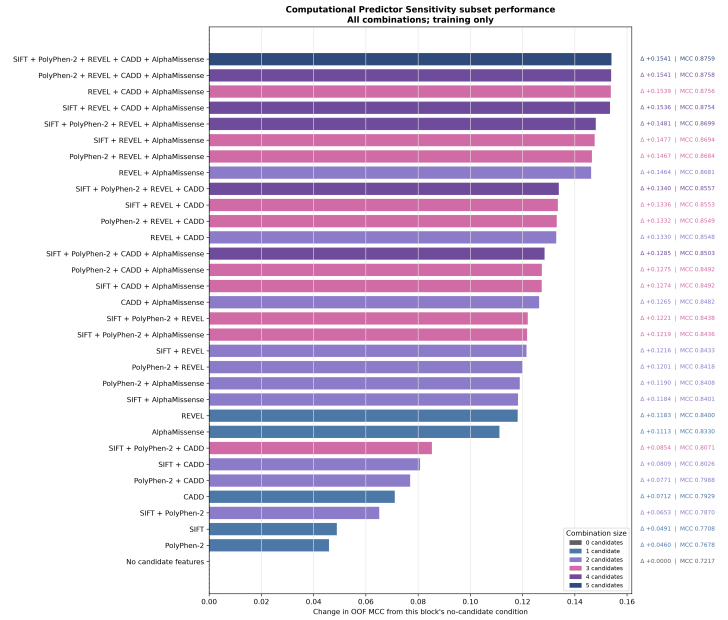



Dominance view for 5.4 population-frequency sensitivity. Bars show exact Shapley/general-dominance contributions to OOF MCC; whiskers show 95% gene-bootstrap intervals. Percentages are normalised within the block and do not imply causality.

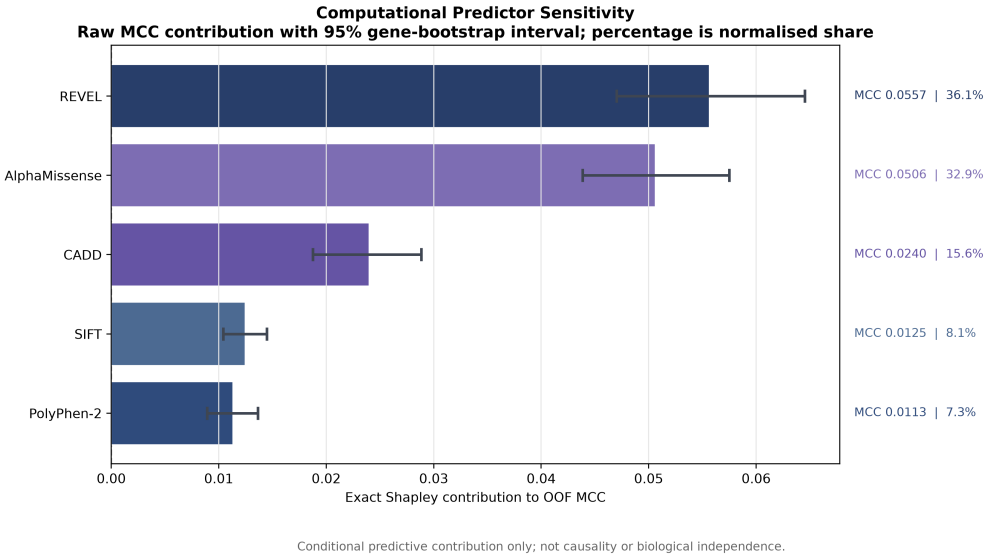
Interpretation. Maximum-population AF alone reached MCC 0.77458 compared with 0.72173 for the no-population background. Adding global AF after maximum-population AF changed MCC by -0.000071, with an interval crossing zero. Maximum-population AF therefore represented the parsimonious sensitivity condition. The block remained sensitivity-only because both fields had 59.75% coverage and class-associated missingness.

# 5.5 Computational-predictor sensitivity

All 32 combinations of five composite predictors were tested on top of the 49-feature primary biological background.




Subset-performance view for 5.5 computational-predictor sensitivity. Bars show the change in pooled OOF MCC relative to the same block's no-candidate condition. Raw MCC values are printed on the figure.




Dominance view for 5.5 computational-predictor sensitivity. Bars show exact Shapley/general-dominance contributions to OOF MCC; whiskers show 95% gene-bootstrap intervals. Percentages are normalised within the block and do not imply causality.

Interpretation. The complete predictor set reached MCC 0.87586 compared with 0.72173 for the biological background. REVEL and AlphaMissense received the largest dominance shares, followed by CADD. The full five-predictor set exceeded REVEL + CADD + AlphaMissense by only 0.000215 MCC, with an interval crossing zero. These results remained diagnostic because predictive gain could not override provenance or circularity governance.

# 6. Group-aware feature selection

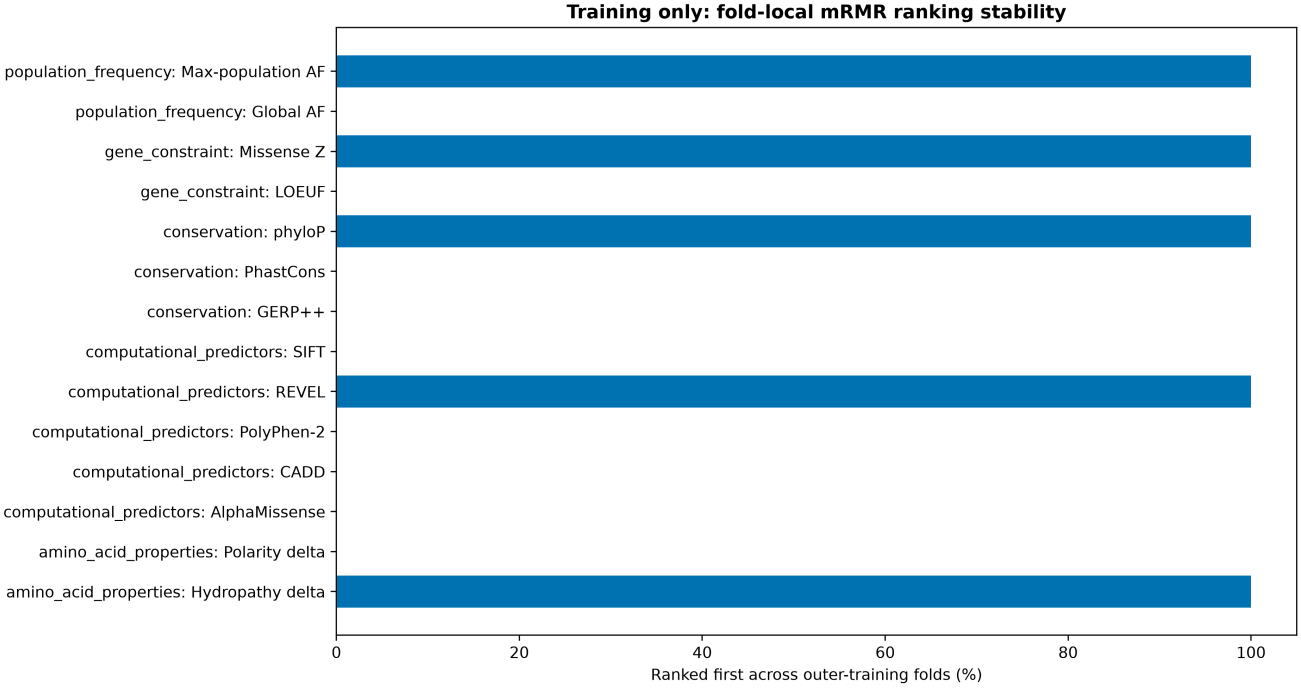
Dominance analysis described how fixed-model performance was distributed across correlated candidates. A second analysis was required to test whether a smaller subset would be chosen consistently when different genes were held out. Three linked procedures were therefore applied.

| Procedure | Question answered | Safeguard |
|---|---|---|
| Fold-local mRMR | Which candidates combine outcome relevance with lower redundancy? | Ranking was calculated only inside each outer-training portion. |
| Nested wrapper selection | Which subset maximises inner gene-grouped OOF MCC? | Four inner gene folds selected the subset; the outer gene fold evaluated it. |
| Fixed-subset outer OOF comparison | How do all predefined subsets perform on unseen outer-fold genes? | Identical outer folds and preprocessing were used for every subset. |
| Paired gene bootstrap | Are full-versus-reduced losses stable across genes? | The same 5,000 gene resamples were applied to paired conditions. |
| Non-inferiority sensitivity | Is a reduced subset likely to lose no more than 0.0025, 0.005 or 0.010 MCC? | Margins were sensitivity values, not post-result automatic thresholds. |

This analysis did not replace dominance analysis. A feature could rank highly in mRMR yet add little to a complete subset, and a subset could achieve the best observed MCC without being selected consistently across outer folds. Agreement across methods strengthened a conclusion; disagreement was retained as uncertainty.

# 7.1 Fold-local mRMR ranking

mRMR balanced target relevance against redundancy with other candidates in the same evidence block. Rank 1 represented the strongest candidate within a block, not the strongest feature in the entire model.




Mean mRMR rank across the five outer-training portions. phyloP, hydropathy, Missense Z, maximum-population AF and REVEL ranked first within their respective blocks.

The ranking supported phyloP as the central conservation score and placed GERP++ third. Hydropathy ranked above polarity. Missense Z ranked above LOEUF in relevance-redundancy screening, although this ranking did not imply that LOEUF could be removed. REVEL ranked first among composite predictors.

# 7.2 Nested subset-selection stability

Every candidate subset was evaluated using four inner gene-grouped folds inside each outer-training portion. The inner winner was then evaluated on the untouched outer gene fold. The figure records how often each subset was selected across the five outer folds.




Outer-fold subset choices. A frequency of 5/5 indicates complete selection stability; lower frequencies indicate that the preferred representation changed with the held-out genes.

The Missense Z + LOEUF pair was selected in all five outer folds. Conservation split between all three scores (3/5) and phyloP + PhastCons (2/5). Amino-acid choices were unstable: polarity alone and the empty candidate set were each selected in 2/5 folds, while the pair was selected once. Population and predictor blocks also showed more than one selected representation.

# 7.3 Non-inferiority sensitivity

The strictest sensitivity question asked whether each reduced subset had at least 95% bootstrap probability of losing no more than 0.0025 MCC relative to the corresponding complete subset. This was not a universal scientific threshold; it was a prespecified decision aid.

**Training only: provisional block-level decision summary**
**Not an authoritative model lock**

| Block | Decision | Provisional subset |
|---|---|---|
| Conservation | inconclusive—requires supervisor decision | all three vs phyloP + PhastCons unresolved |
| Amino Acid Properties | inconclusive—requires supervisor decision | hydropathy/polarity/full alternatives unstable |
| Gene Constraint | retain as grouped representation | Missense Z + LOEUF |
| Population Frequency | sensitivity-only | maximum-population AF sensitivity |
| Computational Predictors | sensitivity-only | REVEL + CADD + AlphaMissense non-inferior diagnostic subset |

Strict-margin probabilities. Purple bars reached the 95% probability criterion. Grey bars did not establish strict-margin non-inferiority.

| Reduced condition | Observed full-minus-reduced loss | P(loss <= 0.0025) | Strict-margin result |
|---|---|---|---|
| phyloP + PhastCons | 0.001315 | 0.8656 | Not established |
| Hydropathy only | 0.000234 | 1.0000 | Established in sensitivity analysis |
| Polarity only | 0.001017 | 0.9046 | Not established |
| Missense Z only | 0.009905 | 0.0352 | Not established |
| LOEUF only | 0.009883 | 0.0728 | Not established |
| Maximum-population AF only | -0.000071 | 1.0000 | Established in sensitivity analysis |
| REVEL + CADD + AlphaMissense | 0.000215 | 0.9998 | Established in sensitivity analysis |

# 8. Integrated interpretation

| Block | Dominance signal | Group-aware signal | Audit-level interpretation |
|---|---|---|---|
| Conservation | phyloP dominated; GERP++ added 0.001315 after phyloP + PhastCons with an interval crossing zero. | All three selected 3/5; pair selected 2/5. Strict 0.0025 non-inferiority was not established. | The committed audit retained uncertainty between the full and parsimonious representations. |
| Amino acid | Small total gain; hydropathy contributed more; incremental polarity gain was uncertain. | Hydropathy ranked first, but outer selections were unstable. Hydropathy alone met the strict sensitivity margin. | No automatic deletion was justified; biological representation remained a review question. |
| Gene constraint | Balanced contributions; removing either produced a supported loss. | The pair was selected 5/5; neither singleton met non-inferiority margins. | Strongest evidence for a grouped two-feature representation. |
| Population | Maximum-population AF carried most useful signal; global AF added no stable increment. | Maximum-population AF ranked first and met strict non-inferiority. | Preferred sensitivity representation, but missingness prevented primary promotion. |
| Predictors | REVEL and AlphaMissense dominated; CADD added value; SIFT and PolyPhen-2 added little to the trio. | The trio met strict non-inferiority, although exact nested subsets varied. | Useful diagnostic comparator only; governance remained unchanged. |

## What the combined evidence supports

- A strong conservation signal was demonstrated, led by phyloP and supported by PhastCons.
- Missense Z and LOEUF showed complementary predictive value as a pair under the fixed audit model.
- Maximum-population AF was the more informative population representation, but incomplete and class-associated coverage remained a serious governance limitation.
- Composite predictors improved discrimination strongly but were not promoted to primary evidence because performance cannot remove methodological or training-data overlap.
- Amino-acid hydropathy and polarity required biological interpretation because statistical selection was not stable across genes.

## 8.1 Limitations and interpretation boundaries

| Boundary | Implication |
|---|---|
| Fixed audit model | Contributions are conditional on balanced logistic regression and may differ under nonlinear model families. |
| MCC at threshold 0.5 | The dominance value function reflects one predefined classification threshold, although ranking and calibration metrics were also recorded. |
| Correlated candidates | Dominance shares distribute shared predictive gain; they do not prove that features are independent. |
| No causality | Higher contribution does not establish a causal biological mechanism. |
| No circularity clearance | Strong performance or low correlation cannot prove absence of training-data or methodological overlap. |
| Training-only uncertainty | All conclusions remain provisional until a configuration is formally locked and confirmed once on validation. |
| Population missingness | Unavailable frequency cannot be interpreted as zero or confirmed rarity. |
| Concept-bottleneck design | These analyses evaluate candidate measurements; they do not by themselves create or validate concept labels. |

The analysis was intentionally conservative. Small or unstable gains were reported as uncertain rather than converted into automatic keep-or-drop rules. Biological meaning, provenance, missingness and interpretability were required alongside predictive performance.

# 9. Output guide and terminology

| Term | Meaning in this audit |
|---|---|
| OOF prediction | A prediction for a training variant produced by a model that was not fitted on that variant's gene fold. |
| MCC | Matthews correlation coefficient calculated from true positives, true negatives, false positives and false negatives. |
| Dominance contribution | Average marginal change in OOF MCC when a feature is added across every subset context. |
| Normalised share | A feature's raw contribution divided by the block's full-minus-empty MCC gain. |
| mRMR | A ranking method balancing target relevance against redundancy among candidate features. |
| Nested selection | Subset choice performed in inner folds and evaluated on an untouched outer gene fold. |
| Gene bootstrap | Uncertainty estimation by resampling genes, retaining all variants belonging to each sampled gene. |
| Non-inferiority sensitivity | Estimated probability that a reduced subset loses no more than a stated MCC margin. |

## Authoritative source artifacts

- 05_dominance_hierarchical_partitioning/ANALYSIS_SPECIFICATION.md
- 05_dominance_hierarchical_partitioning/DOMINANCE_HIERARCHICAL_PARTITIONING_REPORT.md
- 05_dominance_hierarchical_partitioning/subset_performance.csv
- 05_dominance_hierarchical_partitioning/dominance_contributions.csv
- 05_dominance_hierarchical_partitioning/paired_subset_comparisons.csv
- 05_group_aware_feature_selection/ANALYSIS_SPECIFICATION.md
- 05_group_aware_feature_selection/GROUP_AWARE_FEATURE_SELECTION_REPORT.md
- 05_group_aware_feature_selection/mrmr_selection_stability.csv
- 05_group_aware_feature_selection/subset_selection_frequencies.csv
- 05_group_aware_feature_selection/decision_sensitivity_by_margin.csv

All listed paths are relative to outputs/pre_modeling_evidence_audit/. The PDF summarises the committed training-only audit and does not modify the frozen dataset, locked split, authoritative configuration or validation command.